

Performance Analytic

Business Performance Analysis of Kimia Farma for the Period 2020–2023

Kimia Farma – Big Data Analytics

Presented by

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Industrial Engineering graduate with strong analytical and problem-solving skills, specializing in data analytics and data-driven decision making. Skilled in data cleaning, exploratory data analysis (EDA), and visualization using Python, SQL, Excel, and Power BI. Experienced in transforming raw data into actionable insights through academic projects and professional training. Passionate about leveraging data to optimize processes, improve business efficiency, and support strategic decision-making.



Warehouse Operations
(September 2022- May 2023)



Production Data Staff
(May 2023- December 2023)



Production Planning & Inventory Control
(January 2024- December 2024)

Courses and Certification

- **Bootcamp Data Science – Rakamin Academy** | <https://drive.google.com/file/d/17tR2-APIKoFy34jB9MX7jvpORm1iXC9J/view?usp=sharing> | March – August 2025
- **Business Intelligence Analyst Intern – Bank Muamalat** | https://drive.google.com/file/d/1-SH1KuCdPWNe6fFqcbTpvF4ksfeEyPR_/view?usp=sharing | September – October 2025

About Company *kimia farma*

Kimia Farma is one of the largest and oldest pharmaceutical companies in Indonesia. It was established in **1817** during the Dutch colonial period under the name **NV Chemicalien Handle Rathkamp & Co.** After Indonesia's independence, the company was nationalized and became a **state-owned enterprise (SOE)** under the name **PT Kimia Farma (Persero) Tbk.**

The company is engaged in the **pharmaceutical and healthcare industry**, covering a wide range of business sectors, including:

- **Manufacturing** of medicines, traditional herbal products, and raw pharmaceutical materials.
- **Distribution and trading** of pharmaceutical and healthcare products.
- **Retail pharmacy network**, through hundreds of **Kimia Farma pharmacies** across Indonesia.
- **Clinical laboratories, health clinics, and hospitals**, providing direct healthcare services to the public.

Kimia Farma has played a key role in improving **public access to medicines and healthcare services** throughout Indonesia. It is also known for producing **generic and branded drugs** at affordable prices to support national health programs. The company continues to expand and innovate in the fields of **biotechnology, digital health, and cosmetic products**, while also engaging in international markets through exports and global partnerships.

Project Portfolio

Background

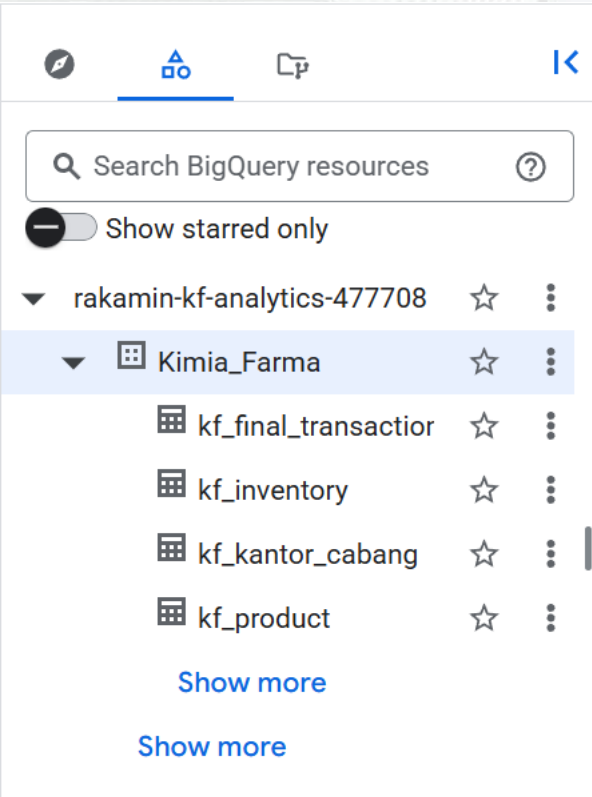
The project is based on aggregating and analyzing multi-source operational data from Kimia Farma, a major pharmaceutical entity. The datasets encompass detailed transaction records, branch information product master data, and inventory details. The core challenge lies in transforming these disparate data silos into a unified, rich analytical table to enable comprehensive business performance evaluation.

Objective

The primary objectives of processing and aggregating these datasets are:

- **To Unify Data for Business Insight:** To consolidate transactional, geographical, and product data into a single, cohesive table in Google BigQuery.
- **To Calculate Key Financial Metrics (KPIs):** To derive crucial financial indicators, specifically **Net Sales** and **Net Profit**, by applying a dynamic gross profit percentage rule based on product price.
- **To Enable Performance Analysis:** To facilitate detailed reporting on sales performance, profitability, customer ratings, and branch efficiency across various products and regional branches (cities/provinces) to support strategic decision-making.

1. Importing Dataset to BigQuery



1. Create New Project “Rakamin-kf-analytic” then create **Kimia_Farma Dataset**.

2. Click **The Kimia_Farma dataset**

3. Click **Create Table**.

4. Select the following:

1. **Source: Upload**

2. **Select file:** Upload the **kf_final_transaction.csv** file

3. **File format: CSV**

4. **Table name:** **kf_final_transaction**

5. **Schema:** Select **Auto detect**

5. Click **Create Table**.

6. Repeat the same steps for:

1. **kf_kantor_cabang.csv**

2. **kf_product.csv**

3. **kf_inventory.csv**

Overview

Details

Tables

Models

Routines

☰ Filter Enter property name or value



Table ID	Type	Create time	Expiration	
kf_final_transaction	Table	November...	None	⋮ ☆
kf_inventory	Table	November...	None	⋮ ☆
kf_kantor_cabang	Table	November...	None	⋮ ☆
kf_product	Table	November...	None	⋮ ☆

2. Tabel Analisa

🔗 rakamin-kf-analytics-477708 / Datasets / Kimia_Farma / Tables / analisa_kimia_farma

analisa_kimia_farma

🔍 Query

Open in ▾

👤 Share ▾

📄 Copy

📷 Snapshot

🗑️ Delete

📤 Export

🔄 Refresh

Schema Details Preview Table Explorer **Preview** Insights Lineage Data Profile Data Quality

Row	transaction_id	date	branch_id	branch_name	kota	provinsi	rating_cabang	custo
2	TRX6978726	2022-05-26	93878	Kimia Farma - Apotek	Semarang	Jawa Tengah	4.5	Suzar
3	TRX4949710	2023-04-02	65948	Kimia Farma - Apotek	Tarakan	Kalimantan Utara	4.4	Kathr
4	TRX9996563	2021-06-01	43007	Kimia Farma - Apotek	Ciamis	Jawa Barat	4.0	Lisa C
5	TRX6539812	2021-01-25	17275	Kimia Farma - Apotek	Banda Aceh	Aceh	3.9	Linda
6	TRX9584952	2021-10-08	59275	Kimia Farma - Apotek	Garut	Jawa Barat	4.6	Katie
7	TRX6565400	2020-01-05	90734	Kimia Farma - Apotek	Sukabumi	Jawa Barat	4.6	Belinc
8	TRX9876181	2023-07-08	47344	Kimia Farma - Apotek	Tanjungpinang	Kepulauan Riau	4.9	Phillip
9	TRX6453359	2021-09-19	12027	Kimia Farma - Apotek	Pangkalpinang	Bangka Belitung	4.8	Patric
10	TRX7851532	2022-04-23	55171	Kimia Farma - Apotek	Ambon	Maluku	5.0	Ambe
11	TRX9076074	2022-12-29	12560	Kimia Farma - Apotek	Padang Sidempuan	Sumatera Utara	4.4	Joani
12	TRX5455288	2023-02-21	31935	Kimia Farma - Apotek	Tangerang	Banten	4.4	Juan

Results per page: 50 ▾ 1 – 50 of 672458 |< < > >|

3. BigQuery Syntax

```
CREATE TABLE Kimia_Farma.analisa_kimia_farma AS
WITH TransformedData AS (
  SELECT
    -- Kolom dari kf.final.transaction (t)
    t.transaction_id,
    t.date,
    t.branch_id,
    t.customer_name,
    t.product_id,
    t.price AS actual_price,
    t.discount_percentage,
    t.rating AS rating_transaksi,

    -- Kolom dari kf.kantor.cabang (b)
    b.branch_name,
    b.kota,
    b.provinsi,
    b.rating AS rating_cabang,

    -- Kolom dari kf.product (p)
    p.product_name,

    -- Kalkulasi persentase_gross_laba berdasarkan actual_price
    (
      CASE
        WHEN t.price <= 50000 THEN 0.10
        WHEN t.price > 50000 AND t.price <= 100000 THEN 0.15
        WHEN t.price > 100000 AND t.price <= 300000 THEN 0.20
        WHEN t.price > 300000 AND t.price <= 500000 THEN 0.25
        ELSE 0.30 -- t.price > 500000
      END
    ) AS persentase_gross_laba
  FROM
    `Kimia_Farma.kf_final.transaction` AS t
  INNER JOIN
    `Kimia_Farma.kf_kantor.cabang` AS b
    ON t.branch_id = b.branch_id
  INNER JOIN
    `Kimia_Farma.kf_product` AS p
    ON t.product_id = p.product_id
)
```

```
SELECT
  transaction_id,
  date,
  branch_id,
  branch_name,
  kota,
  provinsi,
  rating_cabang,
  customer_name,
  product_id,
  product_name,
  actual_price,
  discount_percentage,
  persentase_gross_laba,

  -- Kalkulasi nett_sales: Harga Aktual * (1 - Diskon)
  (actual_price * (1 - discount_percentage)) AS nett_sales,

  -- Kalkulasi nett_profit: Nett Sales * Persentase Laba
  ((actual_price * (1 - discount_percentage)) * persentase_gross_laba) AS nett_profit,

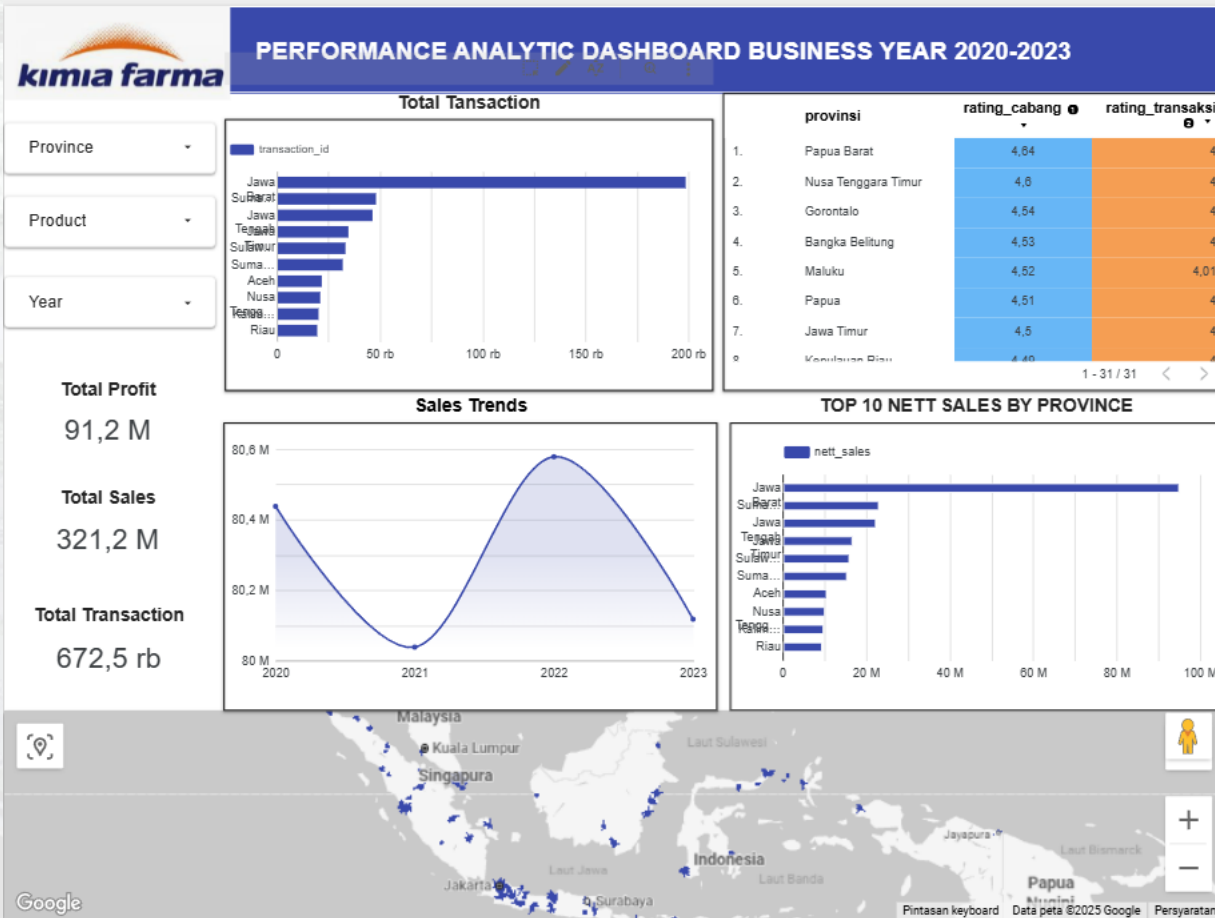
  rating_transaksi
FROM
  TransformedData;
```

Link Github

SQL Query Explanation

- **CREATE TABLE Kimia_Farma.analisa_kimia_farma AS:** This command creates a new table named `analisa_kimia_farma` in the `Kimia_Farma` dataset.
- **WITH TransformedData AS (...):** A Common Table Expression (CTE) is used to perform the necessary joins and calculate the initial columns.
- **INNER JOIN:** The query joins the transaction data (`t`) with the branch office data (`b`) using `branch_id` and the product data (`p`) using `product_id`.
- **persentase_gross_laba:** This is calculated using a **CASE** statement based on the `actual_price` (`t.price`), applying the profit percentage rules you specified.
- **nett_sales:** Calculated as `actual_price * (1 - discount_percentage)`. This is the price paid by the customer after the discount.
- **nett_profit:** Calculated as `nett_sales * persentase_gross_laba`. This represents the estimated profit based on the defined gross profit percentages.
- **rating_cabang** and **rating_transaksi:** The rating columns from their respective tables are aliased to be clear in the final output.

4. Dashboard Performance Analytics



Link Dashboard :

<https://lookerstudio.google.com/reporting/f2abc10d-f850-4702-af36-f42cc7388c72c>

Link Video :

https://drive.google.com/file/d/1PqEgHw0F_hbl3LJuHpK6qED-rwEwOOV/view?usp=sharing

Business Performance Dashboard Analysis (2020-2023)

Key Financial Summary

The overall financial health appears strong, with significant metrics achieved:

- **Total Profit:** \$91.2\$ Million (M)
- **Total Sales:** \$321.2\$ M
- **Total Transaction:** \$672.5\$ Billion (rb)

1. Sales Trends (2020–2023)

The sales trend chart shows a clear, non-linear pattern over the four-year period:

- **2020:** Sales were strong, starting the period at a high level (approx. \$80.4\$ M).
- **2021:** Sales experienced a dip, reaching the lowest point in the trend (approx. \$80.2\$ M).
- **2022:** A significant **peak** was observed, indicating a substantial recovery and the highest sales volume in the period (approaching \$80.6\$ M). This marks the highest growth year.
- **2023:** Sales show a sharp decline from the 2022 peak, ending the period at the lowest sales point (just above \$80\$ M).

2. Total Transaction and Sales by Province

Total Transaction (Volume)

The transaction volume is heavily concentrated in a few provinces:

- **Top 3 Provinces:** Jawa Barat (West Java), Jawa, and Sumatera Selatan (South Sumatra) lead by a significant margin. West Java alone accounts for the highest transaction volume, nearly reaching \$200\$ rb.
- **Lowest Provinces:** Nusa Tenggara and Riau show the lowest transaction volume.

Top 10 Nett Sales by Province (Value)

The distribution of sales value closely mirrors the transaction volume:

- **Top 3 Provinces:** Jawa, Sumatera Selatan, and Jawa Barat are the clear leaders, with sales approaching \$100\$ M.
- **Lower Sales:** Provinces like Nusa Tenggara, Riau, and Aceh contribute the least to net sales.

3. Performance Rating by Province (Branch & Transaction)

This table provides a crucial view into quality and consistency across locations:

- **Top 3 Branch Ratings (rating_cabang):** Papua Barat (West Papua), Nusa Tenggara Timur (East Nusa Tenggara), and Gorontalo hold the highest average branch ratings (above \$4.6\$). This suggests excellent operational quality, service, or customer satisfaction in these smaller regions.
- **Top 3 Transaction Ratings (rating_transaksi):** Most provinces (7 out of 10 listed) have a perfect or near-perfect transaction rating of 4 (or \$4.01\$ for Maluku), indicating a high level of successful transactions company-wide.
- **The Disparity:** There is a **significant disconnect** between the provinces generating the highest Sales/Transactions (Jawa, Sumatera Selatan, Jawa Barat) and those receiving the highest Branch Ratings (Papua Barat, NTT, Gorontalo). The high-volume provinces are not the highest-rated for quality.



Business Recommendations

Based on this analysis, the following strategic recommendations are proposed:

1. Stabilize and Reverse the 2023 Sales Decline

• **Action:** Conduct a deep-dive analysis on the factors that led to the sharp sales peak in **2022** and the subsequent drop in **2023**.

• **Focus Areas:**

- Identify which **Products** or **Promotions** were successful in 2022 and attempt to replicate their success.
- Analyze the market or operational changes in 2023 (e.g., new competitors, inventory issues, macro-economic factors) that caused the slump.

2. Strategic Focus on High-Volume Provinces

- **Action:** Invest in improving operational quality and service in the high-volume provinces (Jawa Barat, Jawa, Sumatera Selatan).
- **Strategy:** The goal should be to close the gap between transaction volume and branch rating. Implement **best practices** from high-rated provinces (Papua Barat, NTT) to improve the **customer experience (rating_cabang)** in West Java and Java. A higher rating in these major markets could significantly boost loyalty and recurring sales.

3. Opportunity in Lower-Performing Provinces

- **Action:** Develop targeted market entry or growth strategies for provinces with low sales/transactions but potentially untapped markets, such as Riau and Nusa Tenggara.
- **Strategy:** Utilize the existing **high Branch Rating** in Nusa Tenggara Timur as a quality anchor. A higher rating suggests a good operational foundation; the focus should be on increasing **marketing, product availability, or distribution channels** to drive transaction volume and net sales.

Thank You

